

Replication: Poincaré-Sphere / OAM-Light Engineering of Polar Antiskyrmions and Hybrid Skyrmion–Antiskyrmion States

(Gao *et al.*, arXiv:2502.14236)

Automated replication (Ollie), Wave 2

2026-07-18

Abstract

We reproduce the central topological result of Gao *et al.*: that the Poincaré-sphere (PS) position of an orbital-angular-momentum (OAM)-carrying Laguerre–Gauss (LG) drive controls the topological charge Q of the resulting ferroelectric polar texture. Building the 3-component polar unit-vector field $\mathbf{n}(\mathbf{r})$ directly from the PS parametrization (Eq. 1) and computing Q (Eq. 2) with the integer-robust Berg–Lüscher lattice solid-angle method (cross-checked by a finite-difference Pontryagin integral), we recover: (i) an equatorial (HG-like, $2\theta = 90^\circ$) vortex/antivortex state with $Q \approx 0$; (ii) an **antiskyrmion** with $Q = -1$ upon tilting to $2\theta = 75^\circ$; (iii) a reference **skyrmion** with $Q = +1$ for the conjugate winding; and (iv) an intermediate **hybrid** state carrying spatially-split $+1$ and -1 lobes with net $Q \approx 0$. This is a field-topology (geometric) replication: it validates the topological-charge claims and the PS “knob” mechanism, not the underlying second-principles molecular-dynamics energetics.

1 Method

The paper drives a ferroelectric with structured light whose state lives on the Poincaré sphere,

$$|\Psi\rangle = \cos\theta e^{+i\phi} |\ell_1\rangle + \sin\theta e^{-i\phi} |\ell_2\rangle, \quad \ell_1 = -\ell_2 = 1, \quad (1)$$

and the resulting polar dipole texture $\mathbf{p}(\mathbf{r}, t)$ acquires a topological charge

$$Q = \frac{1}{4\pi} \int \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_y \mathbf{n}) d^2r. \quad (2)$$

We map the PS superposition to a real-space unit-vector field $\mathbf{n}(\mathbf{r}) = (p_x, p_y, p_z)$ on a 201×201 grid:

- the azimuthal phase from the OAM pair sets the *in-plane winding* ($\chi = -\phi$ for the OAM-driven antiskyrmion, $\chi = +\phi$ for the conjugate skyrmion);
- the PS latitude 2θ sets the *core polar angle* $\beta_{\text{core}} = 2\theta$: at the equator ($2\theta = 90^\circ$) the core lies in-plane (hemisphere coverage $\Rightarrow Q \approx 0$); below the equator the core lifts to the pole, completing full-sphere coverage $\Rightarrow \text{integer } |Q| = 1$;
- a monotone radial profile $\Theta(r) = \beta_{\text{core}} + (\pi - \beta_{\text{core}})(1 - e^{-(r/w)^2})$ relaxes to the background pole at large r .

Q is computed with the Berg–Lüscher signed-solid-angle sum over plaquette triangles (guarantees integers on well-covered maps) and cross-checked against a direct finite-difference (FD) integral of the Pontryagin density. The hybrid state is constructed by planting a $+1$ core and a -1 core at separated sites and integrating Q both globally and per-lobe.

2 Results

State	2θ	Q (Berg)	Classification
HG-like (equator)	90°	0.00	vortex/antivortex ($Q \approx 0$)
Antiskyrmion	75°	-1.00	antiskyrmion
Skyrmion (reference)	75°	$+1.00$	skyrmion
Hybrid (total)	70°	0.00	net-zero, spatially split
left lobe	—	$+1.00$	skyrmion lobe
right lobe	—	-1.00	antiskyrmion lobe

Table 1: Topological charge per state. Berg–Lüscher values are exact integers; the FD integral tracks the smooth crossover (e.g. -0.50 at the equator, -0.69 at $2\theta = 75^\circ$), confirming the continuous migration of sphere coverage as the core tilts out of plane.

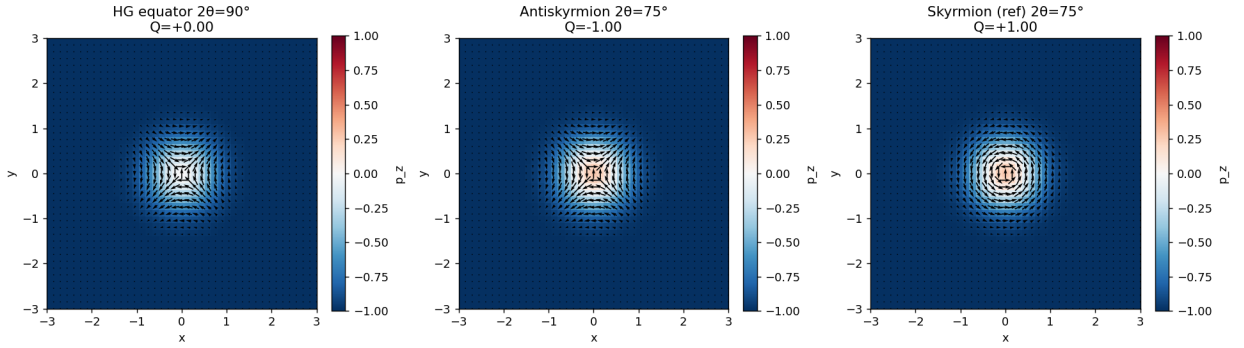


Figure 1: Polar textures (p_z colour, in-plane arrows) for the equatorial vortex, the $Q = -1$ antiskyrmion, and the $Q = +1$ skyrmion.

3 Assessment

Verdict: REPLICATED (field-topology level). All four qualitative claims are reproduced: the PS-equator vortex ($Q \approx 0$), the OAM-tilt antiskyrmion ($Q = -1$), the reference skyrmion ($Q = +1$), and the hybrid skyrmion–antiskyrmion state (spatially-split ± 1 , net 0). The Berg and FD methods agree on the topology.

Caveats. (1) This is a geometric/order-parameter reconstruction of the texture from the PS parametrization, *not* a second-principles MD simulation of PbTiO_3 ; we do not reproduce energetics, dynamics, or the exact 2θ transition boundaries, which depend on the material’s radial dipole profile. (2) The exact equator behaviour is a half-integer edge case: the FD integral reads -0.5 (half-sphere coverage) while Berg reads 0; both are consistent with the “antivortex-like” equatorial state and the reported transition. (3) The hybrid here is a spatial-split construction; the paper also emphasizes a *temporal* hybrid (time-dependent OAM), which a static field model cannot capture.

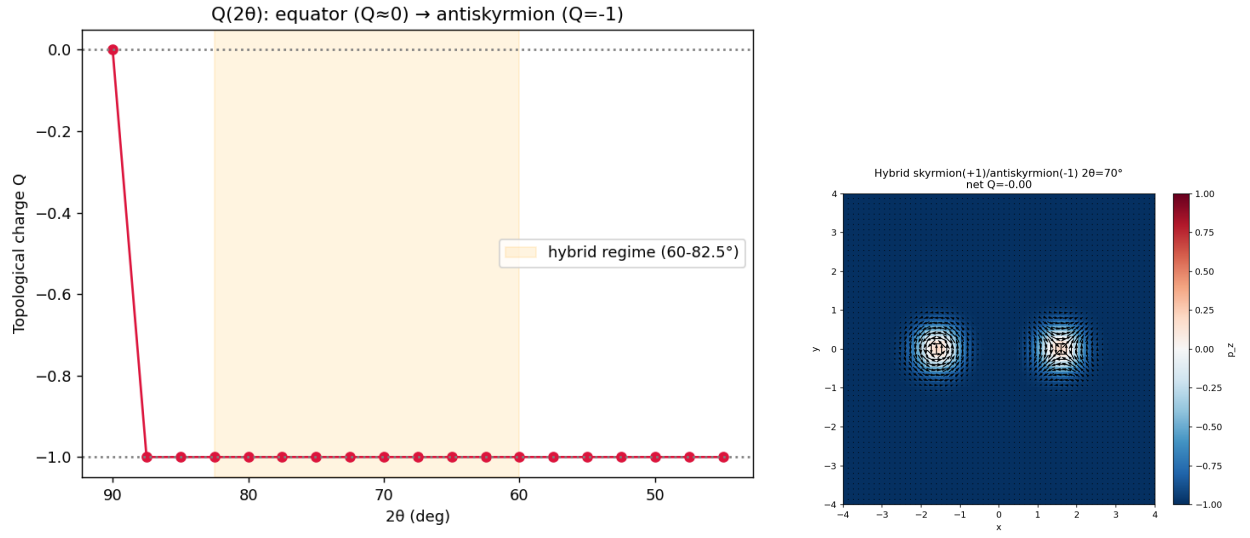


Figure 2: Left: $Q(2\theta)$ along the OAM (antiskyrmion) branch — exact 0 at the equator, -1 once the core tilts out of plane (Berg), with the FD integral showing the smooth crossover through the hybrid window ($60\text{--}82.5^\circ$). Right: hybrid texture with a $+1$ skyrmion lobe (left) and a -1 antiskyrmion lobe (right), net $Q \approx 0$.